A Discriminative Dictionary Learning Approach towards Abnormal Attention Discovery and Autism Spectrum Disorder Identification

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***Abstract*—Autism Spectrum Disorder (ASD) is a group of lifelong neurodevelopmental disorders with complicated causes. A key symptom of ASD patients is their impaired interpersonal communication ability. Recent study shows that face scanning patterns of individuals with ASD are often different from those of typical developing (TD or non-ASD) ones. Such abnormality motivates us to study the feasibility of identifying ASD children based on their face scanning patterns under machine learning. In this paper, we consider using the bag-of-words (BoW) model to encode the face scanning patterns, and propose a novel dictionary learning method based on discriminative mode seeking for better BoW representation. Unlike k-means which is broadly used in conventional BoW models to learn dictionaries, the proposed method captures discriminative information by finding words which maximizes both the purity and coverage of belonging samples within one class. Our work is among one of the few early attempts to directly identify high-functioning ASD children with machine learning. Experiments demonstrate the superior performance of our method with considerable gain over several baselines. Although the proposed work is yet too preliminary to directly replace existing autism diagnostic observation schedules in the clinical practice, it shed light on future applications of machine learning methods in early screening of the disease.**

***Index Terms*—Discriminative Dictionary Learning, Autism Spectrum Disorder, Mode seeking, Machine Learning**

I. Introduction

Autism spectrum disorder (ASD) refers to a group of developmental disorders, including a wide range of symptoms, skills, and levels of disability. Children with ASD often suffer certain lifelong disabilities which cause considerable impacts to their families. While the number of ASD children has risen sharply in recent years, existing ASD diagnostic approaches are both time and labour consuming, causing hinderance to early diagnosis and intervention. Currently, the most widely used assessments include the Autism Diagnostic Observation Schedule-Generic (ADOS-G) [**?**] and its revised version ADOS-2 [**?**]. These diagnostic methods were carefully designed to measure certain behaviors and impairments. Despite their high validity, the accompany and administration of clinically trained professionals are often required. The human-in-loop nature of these tests not only lead to time cost, but also the demand of well controlled protocols and trained professionals.

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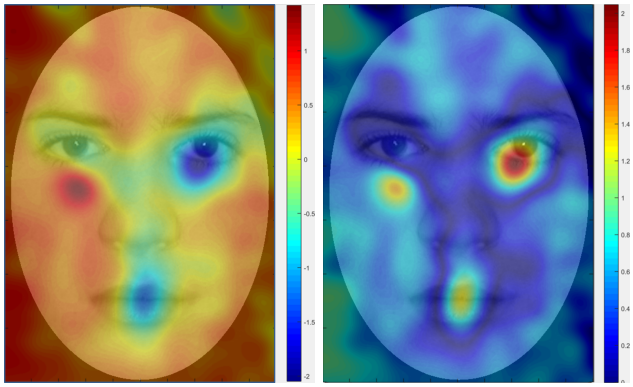

Fig. 1: Heat maps of quality objective values at different face locations. The values encode the different visual preferences between ASD and TD subjects. TD subjects tend to fixate gazes near right eye and mouth, while ASD subjects show certain level of avoidance of eye contact. Our target is to locate these highly different regions, treating them as BoW codebook to encode the gazes. Left: quality objective value. Right: Taking absolute of the left image.

Recent behavioral studies found that ASD individuals show abnormal scanning patterns when looking at faces [**?**]. Similar atypical visual attention is also observed natural static images with general objects [**?**], [**?**]. In these studies, eye gaze captured by eye tracking techniques played a central role in analyzing the ASD behaviors. In the studies, a set of images are displayed on the screen and an eye tracker returns a set of the viewer's eye gaze location (x-y coordinates) on each image. The underlying motivation is that eye gaze patterns, such as the content of viewed objects, fixation durations, viewing frequency of different areas, speed/direction of saccades as well as temporal relations, may encode rich amount of ASD related information. The above studies also motivated recent attempts that use machine learning to identify ASD through abnormal visual attentions [**?**], [**?**].

In this paper, we seek to use learning-based approaches to identify ASD through encoded abnormal visual attention. Our work follows the same identification framework proposed by [**?**] where the gaze coordinates of diagnosed (ASD/non-ASD) subjects [**?**] are encoded into features under the BoW model. Following [**?**], a prediction model is trained by kernel SVM, and evaluated under leave-one-out cross-validation by holding out features from a single subject as test set. An overview of the evaluation protocol is shown in Fig. **??**. Our major contribution lies in proposing a novel dictionary

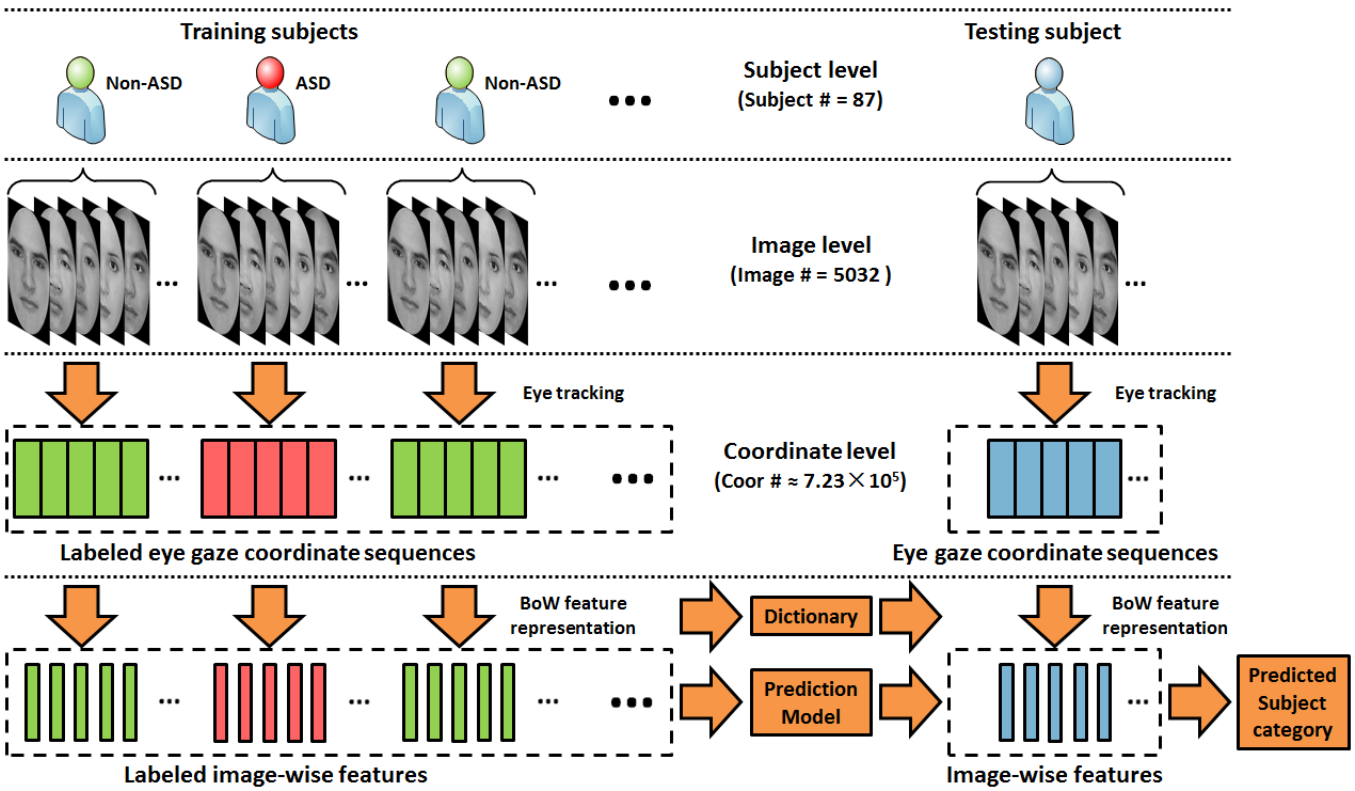

Fig. 2: An overview of the evaluation protocol. Each subject views a set of face images, while the set of eye gaze coordinates on each viewed image are recorded using eye tracking devices. The proposed method encodes the eye gazes at image level with the BoW model.

learning method where high quality codebook words are discriminatively mined. Instead of using k-means to learn words with highly concentrated attentions, our method seeks to locate words that favors concentration difference between ASD and TD gazes. We propose a theoretically unified view towards modeling word quality, by considering **purity** and **coverage** which correspond to inter-class difference and word frequency. We model the quality objective as a product of purity and coverage, and show that such objective can be naturally approximated via kernel density estimation and optimized via discriminative mode seeking. Fig. **??** visualizes the estimated objective values at different locations.

II. Related Work

Our work is related to or partly inspired by a wide variety of previous work, ranging from psychology, psychiatry, behavior analysis to machine learning. Below we give a brief summarization of these work.

A. Psychology, psychiatry and behavior analysis

Using BoW-like orderless representations to model visual attention is partly driven and theoretically supported by considerable amount of work from the communities of psychology, psychiatry and behavior analysis. The area of interest (AOI) is a widely used approach in this community to study human visual attentions [**?**], [**?**]. The spirit of AOI turns out to be similar to the well-known BoW model in machine learning. A brief illustration of the AOI approach from [**?**] is illustrated in Fig. **??**, where the viewed image is manually partitioned into semantically meaningful regions, with the duration of eye gaze (number of gaze coordinates) falling into each region is counted and analyzed. In some sense, these regions can be viewed as pre-defined words in a BoW model. Another method towards analyzing visual attentions is the iMap approach [**?**], where a heat map of visual attention is generated by smoothing eye fixations with Gaussian kernels. Again the concept of iMap coincides with kernel density estimation [**?**], [**?**] and mode seeking [**?**], [**?**]. Recently, multiple studies have reported using AOI and iMap to analyze the abnormal visual attention of ASD [**?**], [**?**], [**?**], [**?**].

B. Machine learning for ASD identification

While the above studies form the theoretical bases of this research, most of them are restricted to statistical studies to discover patterns indicating ASD symptoms. With the fast development and success, machine learning have been introduced to identify ASD. For instance, Kosmicki et al. [**?**] and Duda et al. [**?**] proposed to use machine learning to simplify ADOS/ADOS-G while maintaining high test validity. Some recent autism studies start to address the task of direct ASD prediction [**?**], [**?**], [**?**]. For example, Crippa et al. proposed to classify low-functioning ASD children based on the kinematic analysis of a reach-to-drop task [**?**]. Using machine learning benefits the identification process from two aspects: 1. Replacing human-in-loop operations with machine learning methods makes the identification process much more

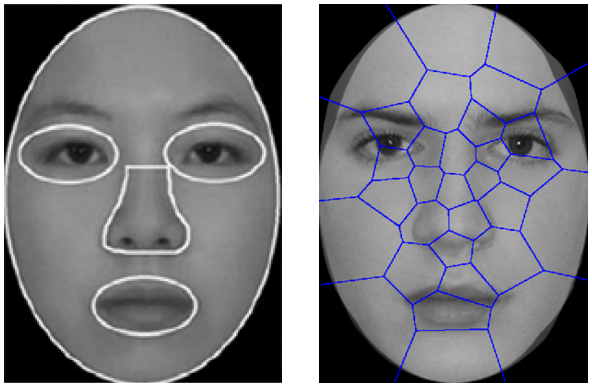

Fig. 3: Illustration of the dictionary words projected onto the viewed image. Left: Partitioned regions as dictionary words in AOI. Right: Regions learned by k-means.

scalable. 2. Learning based methods can generate useful mid-level scores which reduce subjectivity in ASD identification and help better ASD diagnosis.

Motivated by the abnormal visual attentions observed in many ASD studies, Liu et al. [**?**], [**?**] recently proposed an ASD children identification framework with BoW, k-means dictionary learning and SVM, and are so far the most related studies to this work. The difference between AOI and k-means is that the latter is data-driven which does not require manual partitions (See the right image in Fig. **??**). However, k-means finds dictionary words with high sample densities but does not explicitly include discriminative class information. In this work, we follow exactly the same evaluation protocol as presented in [**?**], [**?**]. In particular, the face scanning pattern datasets used in this paper were introduced by [**?**] and [**?**].

Another closely related work is [**?**] where deep network is used to identify ASD by analyzing the viewed contents of test subjects. It is worth mentioning that dataset used by [**?**] is from [**?**] and contains gaze information, which could be used to test the proposed method. However, this dataset is currently not available upon request.

C. General machine learning research

The concept of adding supervised discriminative information to dictionary learning is not new. A number of previous work on sparse dictionary learning have reported improvements over unsupervised dictionary learning when label information is incorporated [**?**], [**?**], [**?**]. While these work presented elegant optimization frameworks for learning sparse dictionaries, their method can not be directly applied to our task given the low dimensional input[1]. Of course, one may argue that higher dimensional inputs can be obtained by sampling the time series of the coordinates in a relatively long window. However, this is much less supported by previous clinical studies in psychology and behavior analysis, and often does not work well since high-order temporal context information along temporal dimension are usually quite random.

The discrete and non-convex nature of k-means makes it even more difficult to incorporate discriminative label information to the codebook learning process similar to sparse dictionary learning. A typical method to heuristically incorporate discriminative information is to learn class-specific codebooks by performing k-means on the class-wise data subsets [**?**]. Other related methods include the descriptive word ranking [**?**] as well as large-margin clustering using SVM and iterative cross-validation [**?**], [**?**].

One related work which partly inspired our proposed framework is [**?**], where the authors discover mid-level image patches with discriminative mode seeking, and formulate the mode seeking process as a constrained optimization problem. While resembling their method in high-level motivations, our work differs from them in several aspects: 1. [**?**] is heavily tailored to handle image classification problems and very high dimensional features. While our method addresses a completely different application. 2. We observe that directly using density ratio as in [**?**] sometimes leads to super large values and undesired learning behaviors when the denominator is small. Instead we consider an alternative purity measurement term where the purity is normalized between 0 and 1. 3. Our model considers the joint optimization of purity and coverage, while the density ratio somewhat discards the coverage information of codebooks. 4. We show that the proposed optimization framework can be elegantly formulated as a supervised mean shift, which is considerably simpler than the constrained optimization in [**?**].

III. Learning Discriminative BoW Dictionary

Dictionary learning presents an important problem in BoW representation as the quality of learned codebook has direct impacts on the quality of represented features. Often, one would hope that the dictionary can encode as much discriminative information as possible, such that the feature coefficients on this dictionary show significant inter-class differences which benefit the classification task. An important question one may ask is: How to quantitatively measure the quality of the given codebook?

A. A unified view towards codebook quality

In [**?**], the authors proposed the concept of **purity** to measure measures how discriminative a dictionary word is, and **coverage** to measure how representative it is. We follow this idea to learn codebooks that have larger values in both terms. Without loss of generality, suppose the codebook is generated by clustering with hard assignment. Given a certain cluster partition $\mathbf{C}$ and the word index w, the purity of the codebook for positive class can be modeled as:

$$p^{+}(w|\mathbf{C}) = \frac{n^{+}(w \mid \mathbf{C})}{n^{+}(w \mid \mathbf{C}) + n^{-}(w \mid \mathbf{C})}, \tag{1}$$

where $n^{+}(w|\mathbf{C})$ and $n^{-}(w|\mathbf{C})$ respectively denotes the numbers of positive and negative samples assigned to word w. Again, note that such measurement differs from the density ratio in [**?**], in the sense that Eq. (**??**) is normalized between 0 and 1. Similarly, the purity of negative class can be defined as:

$$p^{-}(w|\mathbf{C}) = \frac{n^{-}(w \mid \mathbf{C})}{n^{+}(w \mid \mathbf{C}) + n^{-}(w \mid \mathbf{C})}. \tag{2}$$

[1]Our input are x-y coordinates with the dimension of 2.

While it is desirable to increase the dictionary purities for both classes, given the same word, increasing the purity of both positive and negative classes is contradicting to each other. What truly matters is actually the difference of sample numbers and its ratio versus the word size. As a result, we look into the following purity measure:

$$p(w|\mathbf{C}) = \frac{|n^+(w \mid \mathbf{C}) - n^-(w \mid \mathbf{C})|}{n^+(w \mid \mathbf{C}) + n^-(w \mid \mathbf{C})}, \tag{3}$$

which is able to measure the level of purity for both classes with a unified representation. On the other hand, the coverage for positive and negative class can be modeled as:

$$c(w|\mathbf{C}) = n^+(w \mid \mathbf{C}) + n^-(w \mid \mathbf{C}). \tag{4}$$

A codebook ideally should have good purity and coverage simultaneously. A natural way is to treat the product of both benchmarks as the objective, which shares similar spirit to the f-measure[2]. Therefore, the word quality can be estimated as:

$$\begin{aligned} f(w, \mathbf{C}) &\triangleq p(w|\mathbf{C})c(w|\mathbf{C}) \\ &= |n^+(w \mid \mathbf{C}) - n^-(w \mid \mathbf{C})| \end{aligned} \tag{5}$$

The problem of finding a good dictionary word can therefore be formulated as maximizing the quality estimation objective with respect to w and $\mathbf{C}$:

$$\max_{w,\mathbf{C}} f(w, \mathbf{C}) = \mid n^+(w|\mathbf{C}) - n^-(w|\mathbf{C}) \mid \tag{6}$$

B. Approximating with kernel density estimation

Directly optimizing $f(w, \mathbf{C})$ is very difficult since the solution space of $\mathbf{C}$ is huge. the problem is also discrete and non-convex under hard word assignment. However, when the size of each dictionary word is reasonably small enough, a good approximation to $n^+(w|\mathbf{C})$ and $n^-(w|\mathbf{C})$ is the local density estimator centered on each word:

$$\begin{aligned} \hat{p}(\mathbf{x}_w|\mathbf{X}^+) &\propsim n^+(w \mid \mathbf{C}) \\ \hat{p}(\mathbf{x}_w|\mathbf{X}^-) &\propsim n^-(w \mid \mathbf{C}) \end{aligned} \tag{7}$$

where $\mathbf{x}_w$ is the word location in feature space, and $\mathbf{X}^+$/$\mathbf{X}^-$ denote the set of gazes from ASD/TD subjects. In addition, $\hat{p}(\mathbf{x}_w|\mathbf{X}^+)$ is the following kernel density estimator:

$$\begin{aligned} \hat{p}(\mathbf{x}|\mathbf{X}^+) &\triangleq \frac{c_{k,d}}{N^+ h^d} \sum_{i=1}^{N^+} k(\| \frac{\mathbf{x} - \mathbf{x}_i}{h} \|^2) \\ \hat{p}(\mathbf{x}|\mathbf{X}^-) &\triangleq \frac{c_{k,d}}{N^- h^d} \sum_{i=1}^{N^-} k(\| \frac{\mathbf{x} - \mathbf{x}_i}{h} \|^2) \end{aligned} \tag{8}$$

Here, d is the dimension of coordinates. We consider adopting the Gaussian kernel, which gives $c_{k,d} = \pi^{(-d/2)}$. The quality estimation of a word centered at $\mathbf{x}$ in the feature space can therefore be approximated as:

$$f(\mathbf{x}) = |\hat{p}(\mathbf{x}|\mathbf{X}^+) - \hat{p}(\mathbf{x}|\mathbf{X}^-)| \tag{9}$$

[2]One may also consider linear combination but this leads to the weight issue between purity and coverage for the different scales.

C. Locating codebook with dual mode seeking

Eq. (**??**) is a continuous function with respect to $\mathbf{x}$. This allows one to optimize the objective with respect to $\mathbf{x}$ using gradient ascent. Since Eq. (**??**) contains absolute values, we consider the alternative quality estimation objective:

$$g(\mathbf{x}) = \hat{p}(\mathbf{x}|\mathbf{X}^+) - \hat{p}(\mathbf{x}|\mathbf{X}^-) \tag{10}$$

Assuming that the gradient ascent/descent process guarantees the monotonic increasing/decreasing of $g(\mathbf{x})$, we have the following theorems:

Lemma 1: $f(\mathbf{x}) = -g(\mathbf{x}), \forall \mathbf{x} \in \{\mathbf{x}|g(\mathbf{x}) < 0\}$.

Remark: The proof is omitted. **Lemma ??** indicates $f(\mathbf{x})$ is equal to flipping the negative portion of $g(\mathbf{x})$ as positive.

Lemma 2: Gradient ascent on $f(\mathbf{x})$ is equal to gradient ascent on $g(\mathbf{x})$, $\forall \mathbf{x} \in \{\mathbf{x}|g(\mathbf{x}) > 0\}$.

Lemma 3: Gradient ascent on $f(\mathbf{x})$ is equal to gradient descent on $g(\mathbf{x})$, $\forall \mathbf{x} \in \{\mathbf{x}|g(\mathbf{x}) < 0\}$.

Remark: Lemma ?? and **??** can be directly concluded from **Lemma ??**. As a result, performing mode seeking on $f(\mathbf{x})$ can be alternatively done by performing dual gradient ascent/descent on $g(\mathbf{x})$ with respect to the gradient $\nabla g(\mathbf{x})$. To simplify the computation, note that we have:

$$\begin{aligned} \nabla \hat{p}(x \mid \mathbf{X}^+) &= \frac{2}{h^2} \hat{p}(x \mid \mathbf{X}^+)(\mathbf{x}_{m^+} - \mathbf{x}) \\ \nabla \hat{p}(x \mid \mathbf{X}^-) &= \frac{2}{h^2} \hat{p}(x \mid \mathbf{X}^-)(\mathbf{x}_{m^-} - \mathbf{x}) \end{aligned} \tag{11}$$

where $\mathbf{x}_{m^+}$ is the weighted mean of positive data samples weighted by kernels:

$$\mathbf{x}_{m^+} \triangleq \frac{\sum_{i=1}^{n^+} k(\mathbf{x}, \mathbf{x}_i)\mathbf{x}_i}{\sum_{i=1}^{n^+} k(\mathbf{x}, \mathbf{x}_i)} \tag{12}$$

$\mathbf{x}_{m^-}$ is defined similarly. The gradient of objective function is therefore computed as:

$$\begin{aligned} \nabla g(\mathbf{x}) = &\frac{2\lambda}{h^2} \Big[\hat{p}(\mathbf{x}|\mathbf{X}^+)(\mathbf{x}_{m^+} - \mathbf{x}) \\ &- \hat{p}(\mathbf{x} \mid \mathbf{X}^-)(\mathbf{x}_{m^-} - \mathbf{x}) \Big] \end{aligned} \tag{13}$$

The above lemmas indicates the following gradient ascent process to perform dual mode seeking:

1) Estimate $\hat{p}(\mathbf{x}_i|\mathbf{X}^+) - \hat{p}(\mathbf{x}_i|\mathbf{X}^-)$.
2) If $\hat{p}(\mathbf{x}_i|\mathbf{X}^+) - \hat{p}(\mathbf{x}_i|\mathbf{X}^-) > 0$, perform mode seeking with respect to $\nabla g(\mathbf{x}_i)$ until convergence.
3) If $\hat{p}(\mathbf{x}_i|\mathbf{X}^+) - \hat{p}(\mathbf{x}_i|\mathbf{X}^-) < 0$, perform mode seeking with respect to $-\nabla g(\mathbf{x}_i)$ until convergence.

where step 2 and 3 are repeated until convergence.

D. Dual mode seeking as supervised mean shift

We show that in reality one does not need to explicitly flip the sign of $\nabla g(\mathbf{x})$ in order to perform dual mode seeking. Let $y_i \in \{1, -1\}$ indicates the label of $\mathbf{x}_i$, the second term of Eq. **??** can be re-written as:

$$\frac{2\lambda}{h^2} [\sum_{i=1}^{n} y_i k(\mathbf{x}, \mathbf{x}_i)][\frac{\sum_{i=1}^{n} y_i k(\mathbf{x}, \mathbf{x}_i)\mathbf{x}_i}{\sum_{i=1}^{n} y_i k(\mathbf{x}, \mathbf{x}_i)} - \mathbf{x}] \tag{14}$$

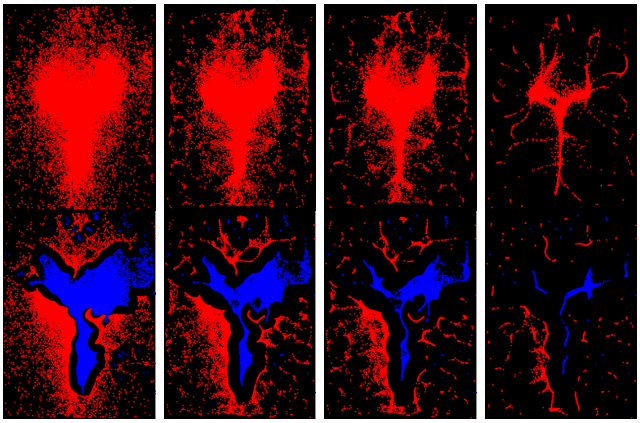

Fig. 4: Visualization of traditional mean shift (top) and proposed method (bottom) at different iterations. For proposed method, red indicates $\hat{p}(\mathbf{x}_i|\mathbf{X}^+) > \hat{p}(\mathbf{x}_i|\mathbf{X}^-) > 0$, while blue indicates $\hat{p}(\mathbf{x}_i|\mathbf{X}^+) < \hat{p}(\mathbf{x}_i|\mathbf{X}^-)$. Left to right: Iteration 1, 5, 10 and 30.

where we have:

$$\sum_{i=1}^{n} y_i k(\mathbf{x}, \mathbf{x}_i) = \hat{p}(\mathbf{x} \mid \mathbf{X}^+) - \hat{p}(\mathbf{x} \mid \mathbf{X}^-) \tag{15}$$

Note that dividing Eq. **??** with $\sum_{i=1}^{n} y_i k(\mathbf{x}, \mathbf{x}_i)$ actually gives a generalized form of mean shift. Also, the sign of $g(\mathbf{x})$ is exactly determined by $\sum_{i=1}^{n} y_i k(\mathbf{x}, \mathbf{x}_i)$. One can cancel the flipping sign of dual mode seeking simply by iteratively shifting with the following mean shift vector:

$$\mathbf{m}(\mathbf{x}) = \frac{\nabla g(\mathbf{x})}{\sum_{i=1}^{n} y_i k(\mathbf{x}, \mathbf{x}_i)} \tag{16}$$

IV. Algorithm Implementation Details

A. Shifting with back tracking line search

Unfortunately, unlike the conventional mean shift, performing gradient ascent with Eq. **??** does not guarantee the monotonic increase of gradient and algorithm convergence, since the sum of kernel weights contains negative terms. This often happens when the densities of positive and negative classes are approximately equal to each other. In this case the denominator of Eq. **??** is very small, leading to relatively large shifting vector or potential numerical issues. This can be practically solved by adaptive step size normalization with respect to the denominator and step size reduction with back tracking line search. Whenever the quality objective value of the next step is not increased, back tracking line search multiplies the current step size with 0.5. This guarantees the monotonic increase of the objective and the algorithm convergence. In practice we observe that mean shift with Eq. **??** works well at most feature space positions, and the need for performing back tracking line search is reduced very fast as the density of one class quickly dominates over another.

B. Computing BoW representation

During training, we use all the data (eye gaze coordinates) from the training set for density estimation. For speed purpose, we sample 1 out of 20 training coordinates to initialize the kernel locations, and perform 30 rounds of mean shifts on these kernels. We keep these settings the same across all our experiments. Once mean shift is finished, we obtain the codebook by performing k-means on the shifted coordinates. We then use the codebook to compute the BoW feature to encode the eye gaze coordinates on each viewed image.

V. Experimental Results

In this section, we report comprehensive evaluations of our method on several datasets.

A. Datasets

We consider two datasets in this paper. The first one, Child Dataset [**?**], includes three groups of participants: 29 Chinese children between 4 to 11 years old with ASD, 29 Chinese TD children matched with the age, and another group of 29 Chinese TD children matched with IQ. And all participants are shown two group of face images: same-race(Asians) faces and other race(Caucasians) faces. All children were diagnosed by experienced clinicians. Note that the dataset used in this paper is exactly the same as the one used by [**?**], but differs from the Child Dataset in [**?**] with additional subjects.

The second one, Adult Dataset, focuses on adolescents and young adults, and is a slightly cleaned up version of the dataset used in [**?**], [**?**], including 19 ASD and 46 non-ASD young adults. As a result, the results between this paper and [**?**] may have certain mismatches, and are not directly comparable.

For both datasets, the eye gaze movements of each person were recorded by a Tobii T60 eye tracker. A set of face images $700 * 500$ are displayed on the screen and eye gaze of each subject is automatically estimated, returning a series of projected coordinates.

B. Evaluation protocol

Following [**?**] and [**?**], we evaluate the proposed method by leave-one-out cross-validation testing, where each subject is consecutively held out for testing while the rest are used for training. By doing this each time we divide the image-level BoW features into two sets: one for testing and the other for training a prediction model. Following [**?**], [**?**], we train an RBF kernel SVM as the prediction model, and predict the test subject score as the mean over the soft SVM prediction scores on images viewed by the test subject. Finally, a global threshold T is set for all testing subjects to obtain the final predictions. For the proposed method and baselines which include the k-means clustering step, we vary the number of clusters as $\{35, 40, 45, 50, 55, 60, 65, 70\}$. For all comparing methods, we also vary the γ and C values in kernel SVM. For fair comparison, the final result of each comparing method is reported as selecting one of the best performing configuration.

C. Evaluation benchmarks

In our experiment, we consider the following benchmarks to quantitatively evaluate the prediction performance:

Area under the curve (AUC): The total area under the ROC

TABLE III: Results on Adult Dataset

	Method	Accuracy	AUC
Adult Dataset	[Liu et al., 2016] [?]	72.31	71.51
	Class K-means	73.85	66.48
	Mean Shift	72.31	68.97
	Class Mean Shift	73.85	72.77
	[Doersch et al., 2016] [?]	**75.39**	73.37
	Proposed	**75.39**	**75.06**

curve versus the whole area. And the ROC curve is a set of (subject-level) true positive rates versus false positive rates obtained by synchronously varied the global threshold T for all testing predictions.

Accuracy: The number of correctly predicted subjects versus the total number of subjects.

Purity: To analyze the level of determinativeness of the dictionaries learned by different methods, we also visualize the dictionary purity profile of comparing methods. **Sensitivity:** To evaluate the actual positives that are correctly identified as such, we compute the sensitivity of each classifier.

D. Baselines

We compare our method with several baselines that are closely related to BoW representations. In particular, we compare our method with dictionary learning methods by k-means [?], class k-means, traditional mean shift, class mean shift, and discriminative mode seeking re-implemented from [?]. For traditional/class mean shift, the bandwidth is cross-validated similar to our method. For class-wise k-means, the number of words learned by positive samples is approximately equal to the number of words learned by negative samples.

E. Visualizing discriminative mode seeking

To visually show how the proposed method works, we visualize the shifted samples at different iterations in Fig. **??**. One could see that the proposed method is able to correctly find both the positive modes and the negative modes belonging to different classes. However, the mean shift method tend to find the most dense regions. For discriminative mode seeking, red color indicates that the initial kernel location belongs to the positive domain, while blue color indicates the negative domain. From left to right: Iteration 1, 5, 10, and 30.

F. Codebook purity analysis

To analyze the determinativeness of codebooks learned by different methods, we also compare the word purities of different methods in Fig. **??**. In particular, we first sort the dictionary words from high to low by the positive class purity, and then plot the purity of the top ranked words. One could see from Fig. **??** that the proposed discriminative mode seeking method tends to have higher purities on than the others. This shows a clear evidence that the proposed method is able to explore the determinative during codebook learning.

G. ROC Curves

The ROC curve of comparing methods on child and adult dataset are shown in Fig. **??**.

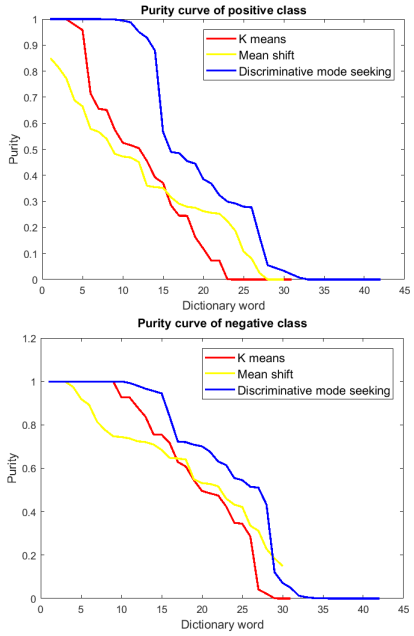

Fig. 5: Positive and negative Purity of different dictionary learning methods, the top contains the purity curves of positive class, while the bottom contains the purity curves of the negative class

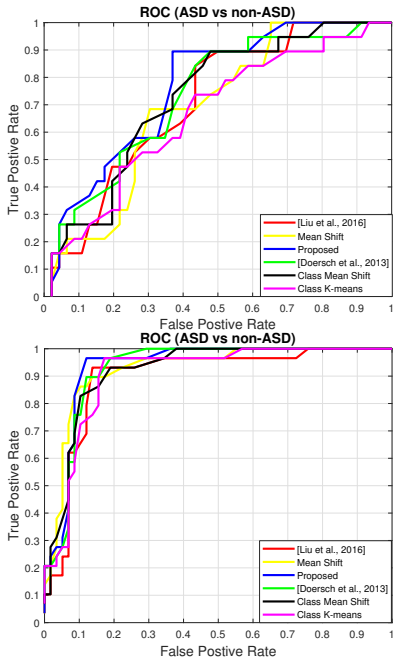

Fig. 6: ROC Curve of comparing methods

H. Result on Child Dataset

Besides experimenting on the complete Child Dataset, we perform our method on the controlled scenario which divides the non-ASD group into to subset(Age-matched and IQ-matched group) following [?], with results shown in Table **??**. Results indicate that overall the proposed performs better than the comparing baselines.

In Child Dataset, each child is shown with face images from two sources: The faces from the same race and faces

TABLE I: Results on Child Dataset with different TD Groups.

Dataset	**All Data**		**ASD v.s. TD-IQ**		**ASD v.s. TD-Age**	
Evaluation Metric	Acc	AUC	Acc	AUC	Acc	AUC
[Liu et al., 2016] [**?**]	88.51	89.63	86.21	88.94	84.48	85.37
Class K-means	87.36	90.79	83.91	84.74	82.76	85.38
Mean Shift	89.66	92.51	87.93	88.59	**87.93**	**88.59**
Class Mean Shift	88.51	92.83	86.21	89.08	86.21	86.87
[Doersch et al.,2013] [**?**]	89.66	92.64	87.93	89.34	**87.93**	87.03
Proposed	**91.95**	**93.40**	**89.66**	**90.96**	**87.93**	87.45

TABLE II: Results on Child Dataset with different subsets of face images."SR" indicates same race (Asian), while "OR" indicates other race (Caucasian).

Dataset	**All Face Images**		**ASD v.s. TD-SR**		**ASD v.s. TD-OR**	
Evaluation Metric	Accuracy	AUC	Accuracy	AUC	Accuracy	AUC
[Liu et al., 2016] [**?**]	88.51	89.63	81.61	82.40	**90.80**	94.41
Class K-means	87.36	90.79	86.21	84.13	89.66	93.40
Mean Shift	89.66	92.51	85.06	**86.50**	90.80	93.34
Class Mean Shift	88.51	92.83	85.06	84.58	89.66	93.87
[Doersch et al., 2016] [**?**]	89.66	92.64	85.06	85.34	89.66	94.03
Proposed	**91.95**	**93.40**	**87.35**	86.27	**90.80**	**94.48**

TABLE IV: Sensitivity of comparing methods.

	Proposed	**[Liu et al., 2016]**	**MeanShift**	**[Doersch et al., 2016]**	**Class MeanShift**	**Class Kmeans**
Adult Dataset	**0.316**	0.158	0.211	0.263	0.263	0.158
Child Dataset	**0.966**	0.931	0.862	0.897	0.828	0.966

from other races. This is another typical setting in psychology study to analyze the ASD behavior. Following this setting, we subdivide our dataset into two subsets, and conduct the same evaluation. Table **??** shows the results of the proposed method and comparing baselines on the Child Dataset.

I. Result on the Adult Dataset

Table **??** shows the results on the Adult Dataset. Again one could see that the proposed method out-performs all comparing baselines with a significant margin.

J. Sensitivity Analysis

We first analyze the sensitivities of comparing methods on adult and child dataset. From Table **??** we can find that the proposed method outperform the comparing methods.

VI. Concluding Remarks

In this paper, we proposed a novel dictionary learning method based on discriminative mode seeking. Our method incorporates label information and can automatically mine discriminative words through supervised mean shift. We also give detailed motivation, intuition, as well as theoretical justification of the proposed method. Experimental results indicate that our machine learning model gives significantly performance over several widely used visual attention encoding methods. Our method can be extended to other features as well. For example we could apply the same codebook learning and BoW representations to motion features and short coordinate sequences in order to incorporate short temporal and higher order information. This will be detailedly addressed in our future work.

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Michael Shell Biography text here.

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Jane Doe Biography text here.